Sungwoo Park

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RESEARCH INTERESTS

My research focuses on traffic management approaches for safe, efficient, and scalable UTM and AAM operations. In particular, I investigate decentralized traffic management, multi-agent conflict resolution, path planning, and GIS-based airspace network design for large-scale operations through game theory, optimization, and reinforcement learning. I am interested in extending these approaches to emerging challenges in autonomous aerial systems.

EDUCATION

Korea Aerospace University Mar. 2025 – Present
Master of Science – MS, Department of Advanced Air Transportation Goyang, South Korea

- **Advisor:** Dr. Sang Hyun Kim
- **Grade:** 4.2 / 4.5
- **Key Courses:** Optimization Theory and Applications, Introduction to Advanced Artificial Neural Networks, Machine Learning with Python, Simulation-Based Disaggregate Choice Modeling

Korea Aerospace University Mar. 2019 – Feb. 2025
Bachelor of Science – BS, Air Transport, Transportation, and Logistics Goyang, South Korea

- **Grade:** 4.08 / 4.5
- **Focus:** Aviation Regulations, Aviation Meteorology, Airport Planning, Air Navigation Facilities and Air Navigation, Aeronautical Information, Aeronautical Communications, ATC Training 1 and 2, Operation Research 1 and 2

PUBLICATIONS AND MANUSCRIPTS

Journal Manuscripts

J1. Ji, S., **Sungwoo Park**, Kim, S. H., "Risk-Based Airspace Capacity Assessment for Small Unmanned Aircraft in Urban Environments."
Transportation Research Part A: Policy and Practice, under review, 2026.

J2. **Sungwoo Park**, S. H. Kim, "Game-Theoretic Approaches to Decentralized UAS Traffic Management."
IEEE Transactions on Intelligent Transportation Systems, manuscript in preparation for submission, 2026.

Conference Publications and Presentations

C1. **Sungwoo Park** and S. H. Kim, "Multilateral Negotiation Among Traffic Management Service Providers."
AIAA AVIATION 2026 Forum, available at [here].

C2. B. Kang, **Sungwoo Park** and S. H. Kim, "Analysis of Available Airspace for Drones in Airport Control Zone."
The Korean Society for Aeronautical and Space Sciences 2025 Fall Conference, available at [here].

C3. **Sungwoo Park**, H. J. Jin, and S. H. Kim, "Ground Risk Analysis for Urban Low Altitude Operations."
The Korean Society for Aeronautical and Space Sciences 2024 Spring Conference, available at [here].

HONORS AND ACADEMIC ACTIVITIES

H1. First Prize, 2023 Air Transportation Capstone Design II, Korea Aerospace University.

H2. Third Prize, 2023 Korea Student UAM Olympiad, Korea Transport Institute.

H3. Second Prize, 2022 Korea Airports Corporation Student Thesis Competition, Korea Airports Corporation.

CERTIFICATIONS AND LICENSES

L1. Pilot Certificate for Ultra-Light Vehicles, Ministry of Land, Infrastructure and Transport, South Korea, 2022.

L2. Machine Learning with Python-From Linear Models to Deep Learning, edX, 2025

TECHNICAL SKILLS

Programming: Python, MATLAB, QGIS, LaTeX
Research Methods: Game theory, Optimization, Simulation
Languages: Korean (Native); English (TOEFL iBT: 5.0 / 6.0)